

AI-Powered Enterprise Email Intelligence

MailMind

Technical Architecture & Developer Documentation

Version	v2.0.0
Date	2025
Stack	FastAPI · Next.js 15 · LangGraph · Azure OpenAI · Supabase
Team	Radiant CodeX
License	MIT

This document provides a comprehensive technical reference for the MailMind codebase — covering system architecture, file structure, per-file API documentation, LangGraph pipeline design, database schema, and deployment infrastructure.

Executive Summary

MailMind is an AI-powered enterprise email triage and drafting platform built for busy executives who receive hundreds of emails per day. It automatically scores every incoming email across five contextual axes, extracts calendar commitments, detects conflicts, retrieves relevant precedents, and generates personalised draft replies — all in under 3 seconds.

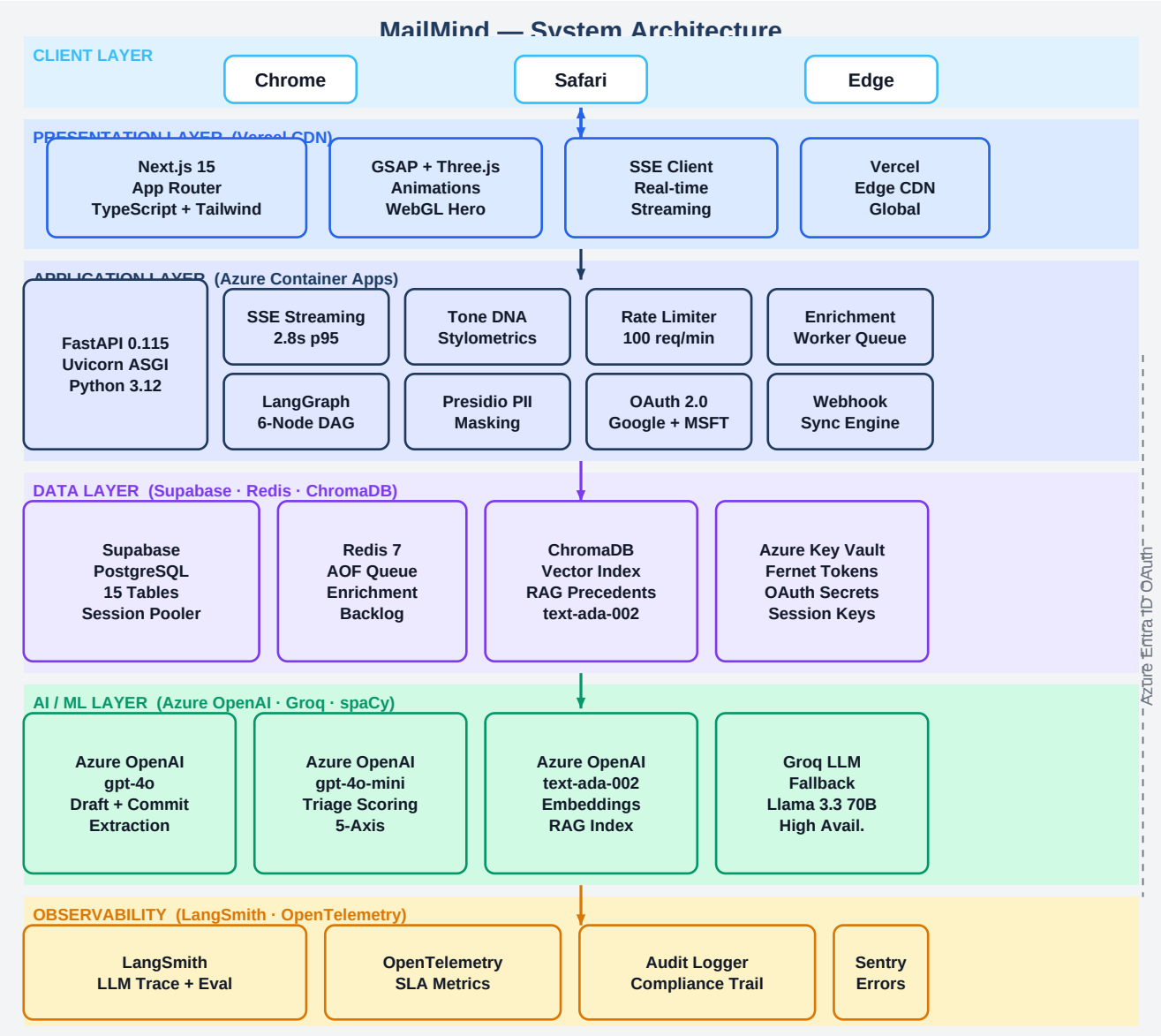
Key Feature	Description	Tech
5-Axis Triage	Scores emails on deadline urgency, sender authority, sentiment, action type, thread decay	GPT-4o-mini
Tone DNA	Learns writing style from 50+ sent emails (8 stylometric features)	spaCy + custom NLP
RAG Drafting	Retrieves 3 similar past emails and injects context into draft generation	ChromaDB + Ada-002
PII Masking	Presidio + spaCy reversible token masking before LLM calls	Microsoft Presidio
Commitment Extraction	Extracts action items with deadlines + calendar conflict detection	GPT-4o structured
Mailbox Mirror	Server-side delta sync (cursor-based) + Graph/Gmail webhooks	Graph API + Pub/Sub
Multi-Account	UUID-based user identity, multiple Gmail/Outlook accounts per user	PostgreSQL
Demo Login	One-click seeded demo with 10 emails, full triage cache, VP persona	seed_demo_account.py

Technology Stack

Backend	Frontend	AI / ML	Infrastructure
- FastAPI 0.115 - Python 3.12 - LangGraph 0.2 - LangChain 0.3 - SQLAlchemy 2.0	- Next.js 15 - TypeScript 5 - Tailwind CSS 3 - GSAP 3 - Three.js	- Azure OpenAI gpt-4o - gpt-4o-mini (triage) - text-embedding-ada-002 - Groq Llama 3.3 - spaCy en_core_web_sm	- Azure Container Apps - Azure Key Vault - Supabase PostgreSQL - Redis 7 AOF - ChromaDB (local)

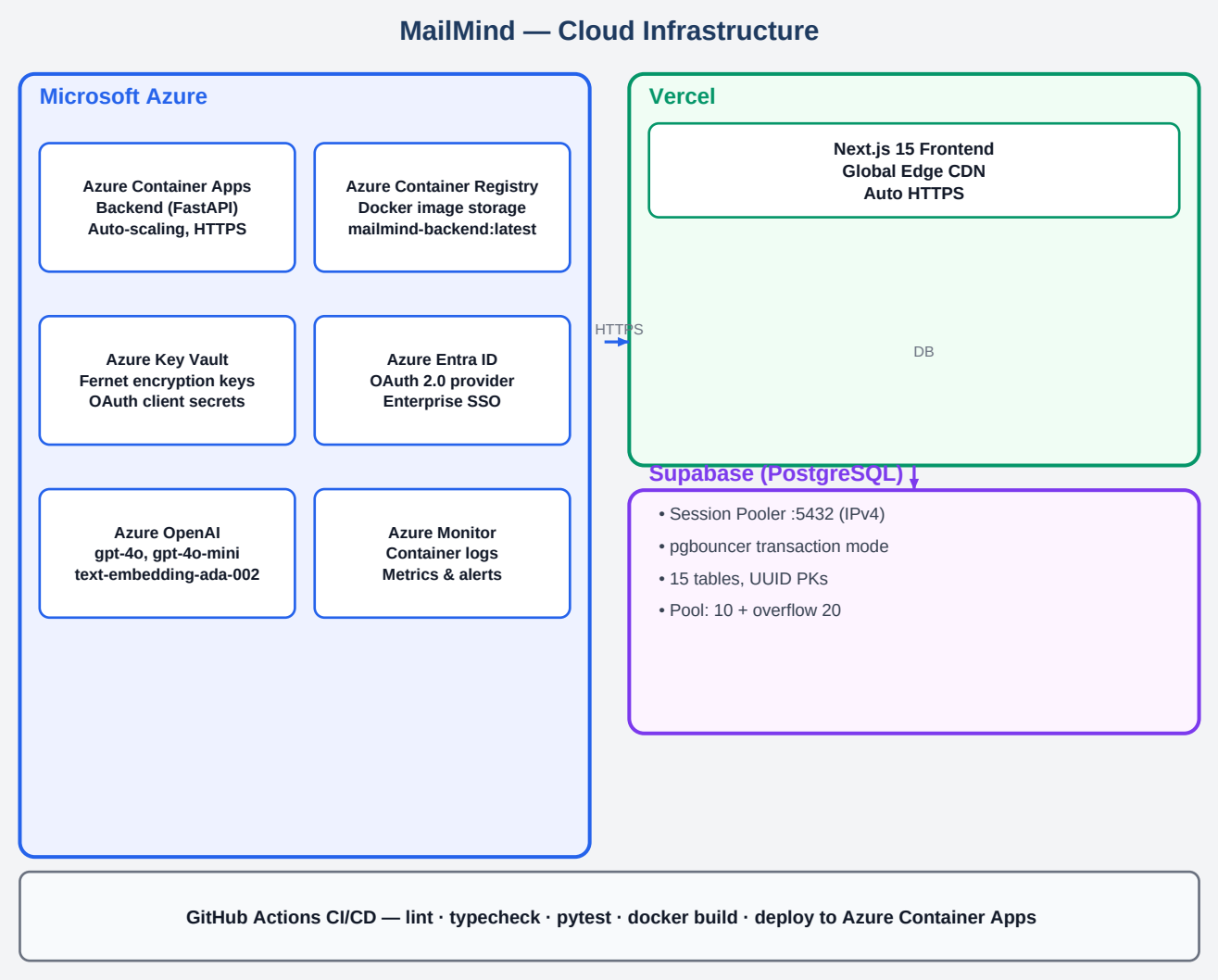
System Architecture

MailMind follows a layered architecture: Next.js frontend on Vercel, FastAPI backend on Azure Container Apps, and separate data/AI/observability layers. All traffic is HTTPS; OAuth tokens are Fernet-AES-128 encrypted at rest.



Cloud Infrastructure

All Azure resources live in a single resource group. Container Apps handle auto-scaling; Key Vault provides secrets management. GitHub Actions builds and deploys on every push to main.



LangGraph Pipeline Architecture

MailMind uses a 6-node LangGraph DAG (Directed Acyclic Graph) to process each email. After ingestion and PII masking, nodes 2–4 (triage, commitment, RAG) run in parallel using `asyncio.gather()`, reducing latency by ~60%. Calendar conflict detection and the approval gate run sequentially after the parallel phase.

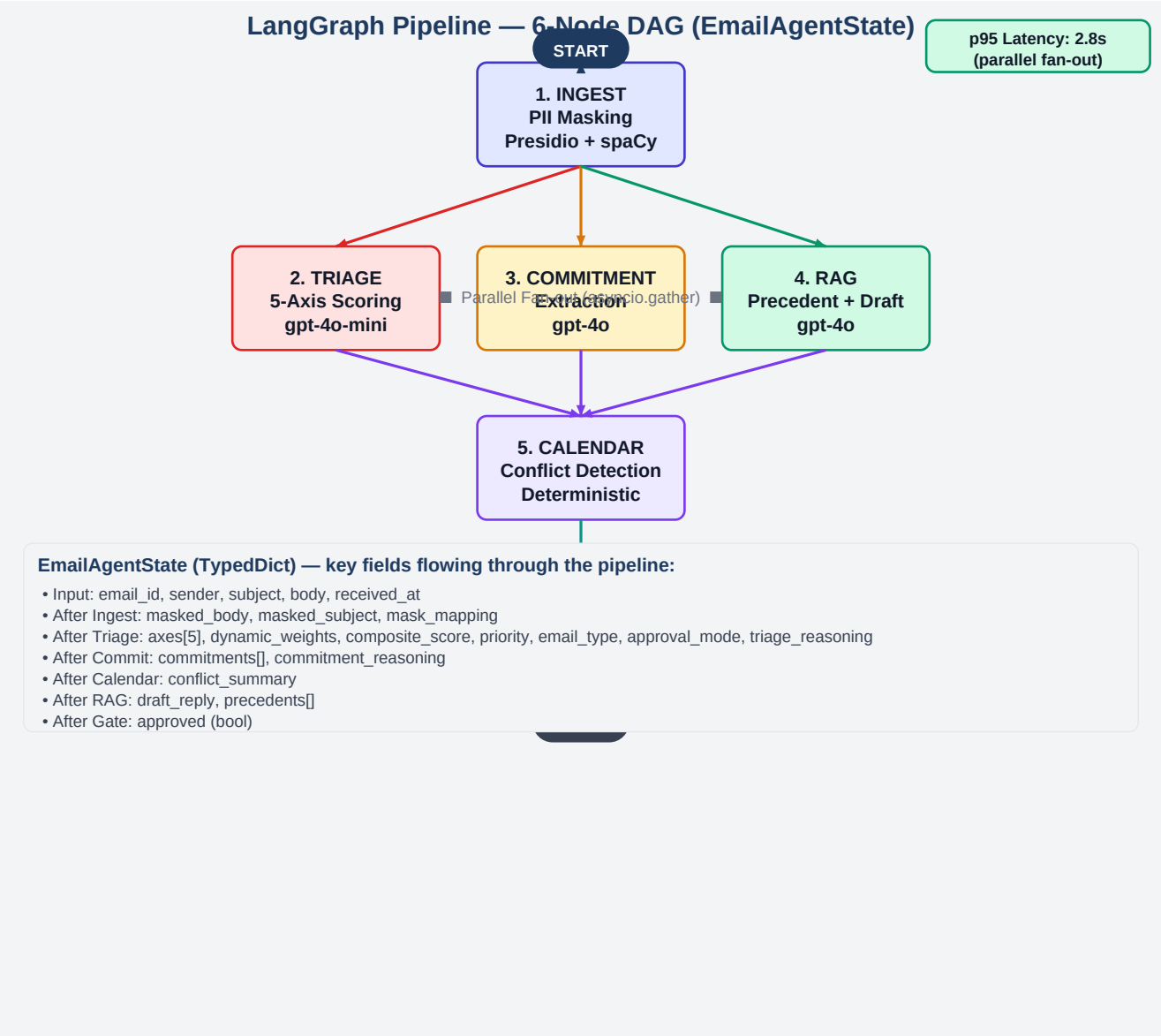


Fig 1: The six pipeline nodes, parallel fan-out edges, and EmailAgentState fields. All nodes share a single TypedDict state object that accumulates results.

OAuth Authentication Flow

MailMind supports Google and Microsoft (Entra ID) OAuth 2.0. After the consent redirect, the backend exchanges the code for tokens, Fernet-encrypts them at rest, upserts the user/account rows, creates a session with SHA-256 hashed token, and returns an HttpOnly cookie (SameSite=Lax).

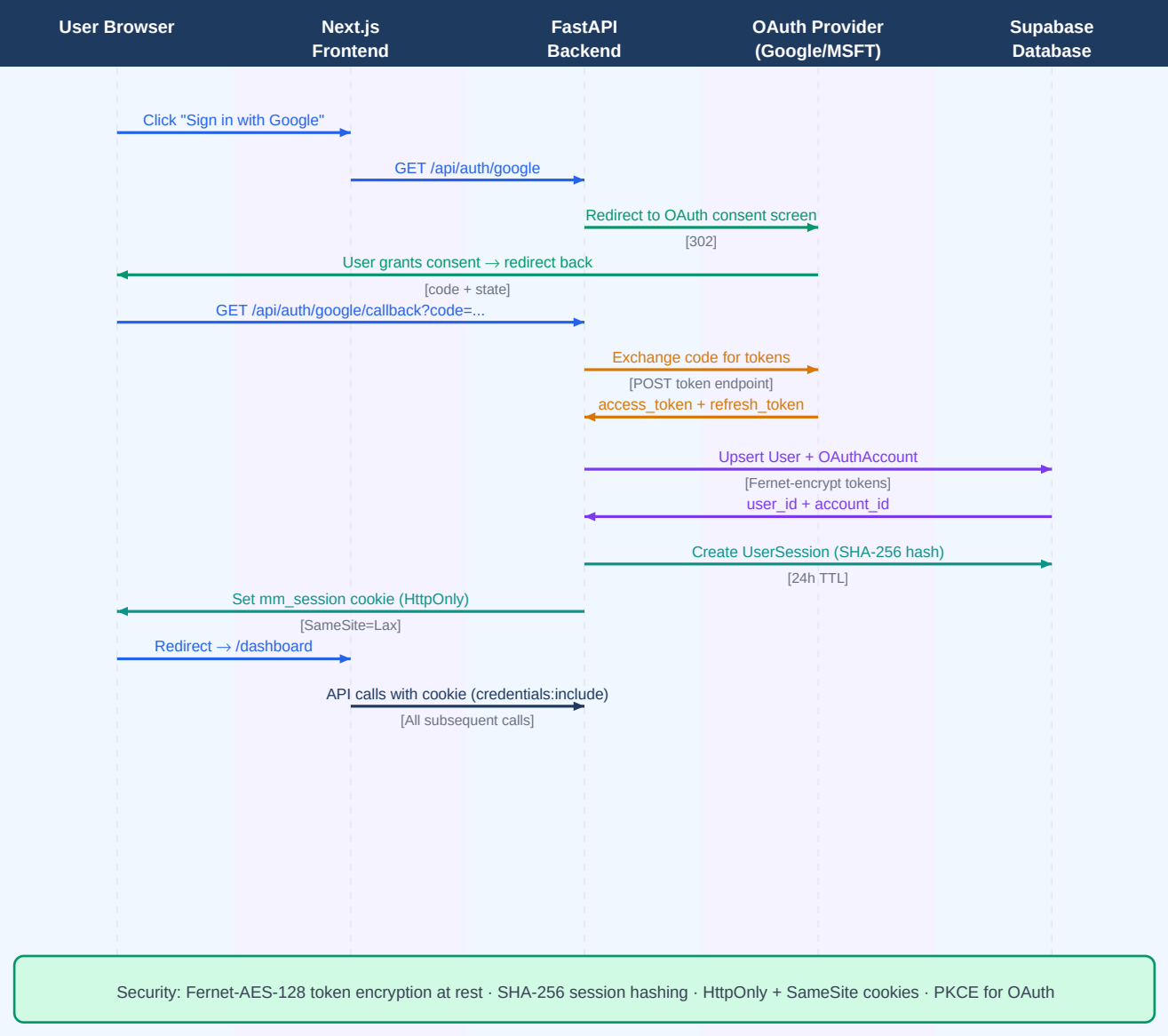
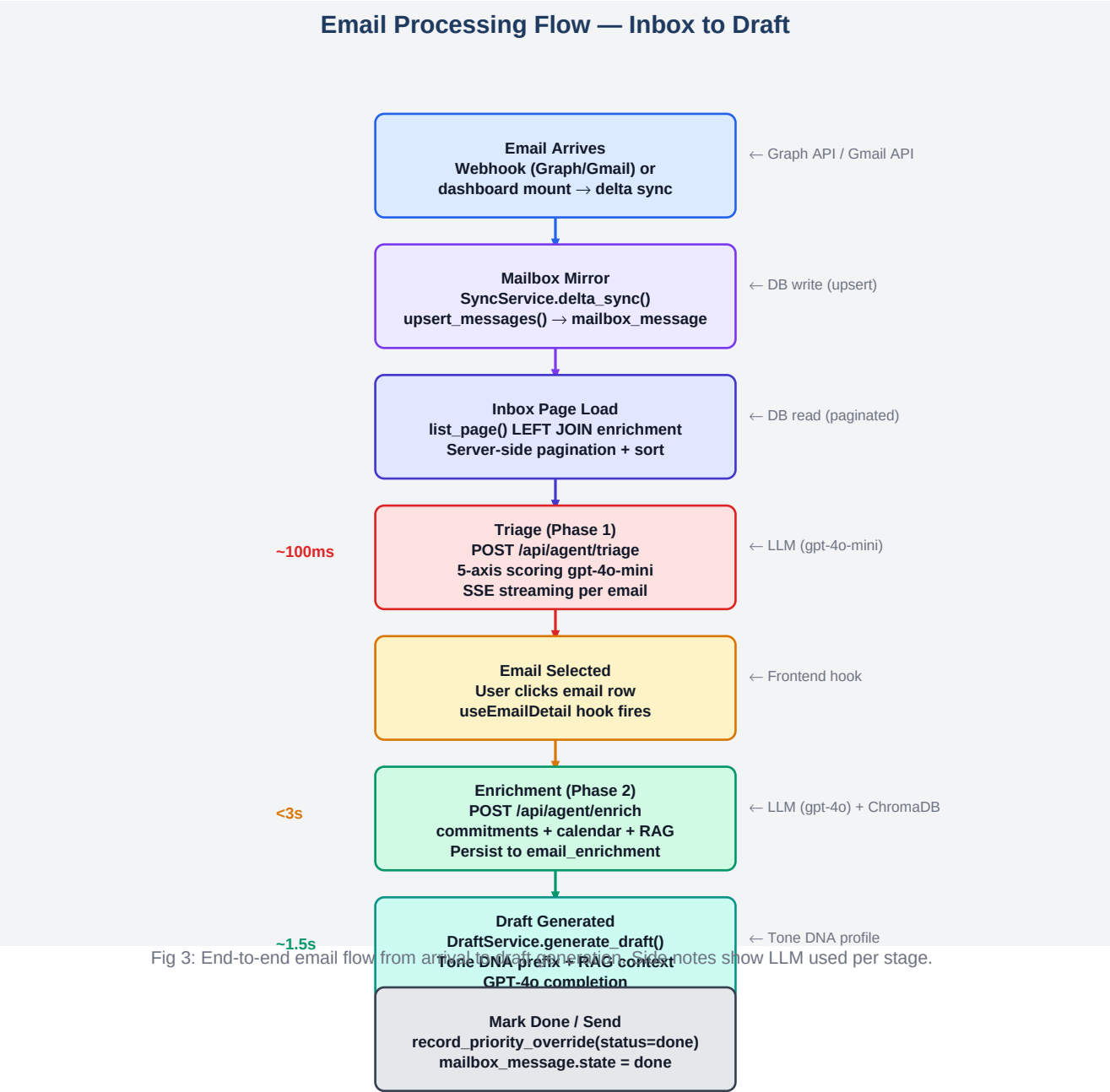


Fig 2: Swim-lane OAuth flow. Tokens encrypted with Fernet-AES-128; sessions hashed with SHA-256.

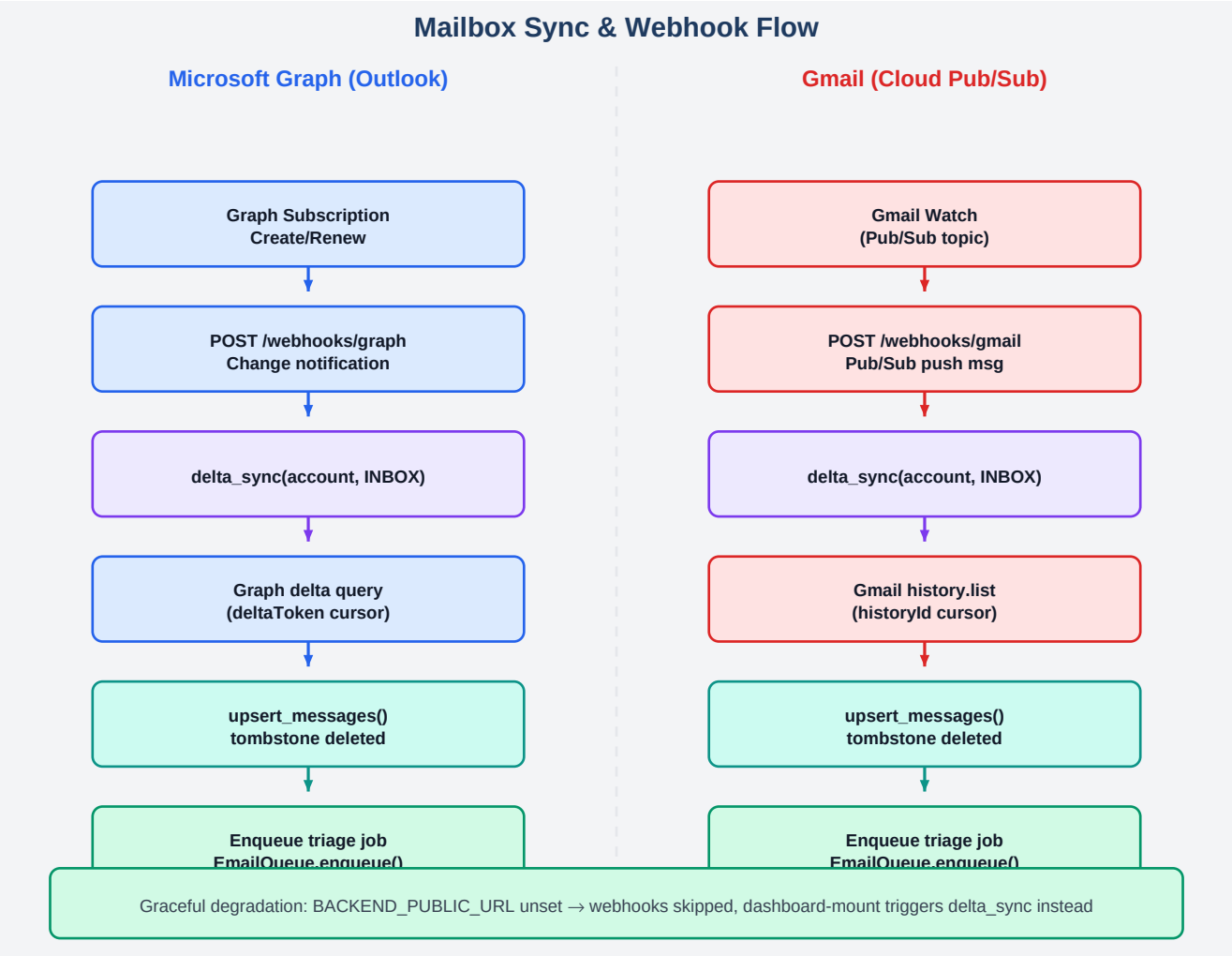
Email Processing Flow

Email processing is split into two user-facing phases. Phase 1 (triage) runs immediately on inbox load and populates priority badges within ~100ms per email using SSE streaming. Phase 2 (enrichment) runs when an email is opened, fetching commitments, calendar conflicts, RAG precedents, and a draft reply.



Mailbox Sync & Webhook Flow

MailMind maintains a server-side mailbox mirror (MailboxMessage table) synced via Microsoft Graph delta queries and Gmail history.list. Webhooks trigger near-real-time delta syncs. If BACKEND_PUBLIC_URL is unset, the system degrades gracefully to on-mount triggered syncs.



Backend File Documentation

This section documents every Python file in the backend, including its purpose, key classes, and all public functions with their parameters and descriptions.

Entry Point & Application Bootstrap

[backend/app/main.py](#)

FastAPI application factory. Configures structured logging, registers ASGI middleware (SessionContext → SecurityHeaders → RateLimit → CORS), mounts all routers, and runs startup tasks via the lifespan context manager.

Method / Function	Parameters	Description
<code>_configure_logging()</code>	→ None	Sets up structured logging with a dedicated mailmind.audit logger. Attaches a FileHandler when AUDIT_LOG_FILE env var is set for an isolated audit sink.
<code>lifespan(app)</code>	app: FastAPI → AsyncGenerator	Startup: creates EmailQueue, calls init_db(), ensures ChromaDB directory, pre-warms Presidio/spaCy (eliminates 4s cold-start), checks LangSmith status.
<code>app = FastAPI(...)</code>	—	Creates FastAPI instance with title, description, version 2.0.0, and lifespan handler.
<code>init_observability(app)</code>	app: FastAPI → None	Registers global exception handlers and optional Sentry integration (from observability.py).
<code>app.add_middleware(...)</code>	—	Middleware stack (reverse order): RateLimit → SecurityHeaders → SessionContext → CORS.
<code>app.include_router(...)</code>	—	Mounts: main router, agent, demo, monitoring, feedback, waitlist, pii, compliance, sync routers.

Configuration

[backend/app/config/settings.py](#)

Single Settings class that reads all environment variables. Loaded once at import time; dotenv first, then Azure Key Vault takes precedence.

Method / Function	Parameters	Description
<code>_bool_env(name, default)</code>	name: str, default: bool → bool	Parses boolean env vars — accepts "1", "true", "yes".
<code>Settings.bootstrap_allowed_set</code>	→ set[str]	Property — lowercase set of always-allowed owner emails (never locked out by waitlist).
<code>Settings.is_production</code>	→ bool	Property — True when APP_ENV == "production".
<code>Settings.persistence_enabled</code>	→ bool	Property — True when DATABASE_URL is non-empty.
<code>Settings.azure_openai_base_endpoint</code>	→ str	Property — strips deployment path from Azure URL to get scheme+host only.

[backend/app/config/keyvault.py](#)

Azure Key Vault integration. Loads secrets into os.environ at startup so downstream code reads them like normal env vars.

Method / Function	Parameters	Description
<code>load_keyvault_into_env()</code>	→ None	Reads AZURE_KEYVAULT_URL; if set, fetches all secrets and writes to os.environ. Requires Azure Managed Identity or AZURE_CLIENT_* credentials.

ASGI Middleware

[backend/app/middleware.py](#)

Three custom ASGI middleware classes stacked on the FastAPI application.

Method / Function	Parameters	Description
<code>SecurityHeadersMiddleware.__call__</code>	<code>request, call_next</code>	Injects: X-Content-Type-Options, X-Frame-Options, Referrer-Policy, Permissions-Policy, and HSTS (when <code>HSTS_ENABLED=true</code> and HTTPS).
<code>RateLimitMiddleware.__call__</code>	<code>request, call_next</code>	Fixed-window per-IP rate limiter (default 100 req/min). Exempts <code>/api/health</code> , <code>/api/ready</code> , and <code>OPTIONS</code> . Returns HTTP 429 with <code>Retry-After</code> header.
<code>SessionContextMiddleware.__call__</code>	<code>request, call_next</code>	Binds a <code>ContextVar</code> (<code>current_session</code>) before the request and clears it after. Downstream code calls <code>get_current_session()</code> instead of passing sessions.

Database Base

[backend/app/db/base.py](#)

SQLAlchemy engine factory and session maker. Configures pool parameters from settings.

Method / Function	Parameters	Description
<code>create_engine(...)</code>	—	Creates async-compatible SQLAlchemy engine with <code>pool_size</code> , <code>max_overflow</code> , <code>pool_timeout</code> , <code>pool_pre_ping</code> , <code>pool_recycle</code> from Settings.
<code>SessionLocal</code>	—	sessionmaker factory; use as context manager or via FastAPI dependency.
<code>init_db()</code>	→ None	Calls <code>Base.metadata.create_all()</code> — creates all ORM tables on first run. No-op when <code>DATABASE_URL</code> is empty.

ORM Models

[backend/app/db/models.py](#)

All 15 SQLAlchemy ORM classes (tables). UUID primary keys throughout. Defines relationships, unique constraints, and JSON columns.

Class (Table)	Description
<code>User</code>	Core identity. UUID PK. One user can have multiple <code>OAuthAccounts</code> .
<code>OAuthAccount</code>	Connected email account (Gmail/Outlook). Fernet-encrypted tokens. <code>UniqueConstraint(provider, provider_account_id)</code> . Links to <code>User</code> .
<code>UserSession</code>	Short-lived session (24h). <code>token_hash = SHA-256(secret + raw_token)</code> . Links to <code>User</code> .
<code>Device</code>	Trusted browser fingerprint for Quick Login. Links to <code>User</code> .
<code>QuickLoginToken</code>	7-day auto-resume token. status: <code>ACTIVE</code> / <code>LOGGED_OUT</code> / <code>REVOKED</code> / <code>EXPIRED</code> .
<code>EmailEnrichment</code>	Triage + enrichment results. JSON columns: <code>axes</code> , <code>dynamic_weights</code> , <code>commitments</code> , <code>precedents</code> . status: <code>pending</code> / <code>done</code> / <code>error</code> .
<code>AuditLog</code>	Append-only compliance trail. No raw PII. action, actor, details (JSON).
<code>ToneProfile</code>	Stylometric profile per account. profile (JSON 8 features). One row per account.
<code>Feedback</code>	Product feedback. rating, category, message, role.

TriagePriorityOverride	User priority feedback loop. original vs override priority per email.
ProcessingMetric	SLA latency per stage. duration_ms, success, sla_met.
MailboxMessage	Server-side envelope mirror. state: active/done/deleted. sender, subject, snippet, received_at, is_read, is_starred.
MailboxSyncState	Delta cursor per account+folder. delta_cursor, backfill_done, last_synced_at, message_count.
GraphSubscription	Microsoft Graph webhook subscription lifecycle. provider_sub_id, client_state, expires_at.
Waitlist	Private-beta access control. status: pending/approved/rejected. source: signup/login.

Enrichment Repository

[backend/app/db/repository.py](#)

Data access layer for email_enrichment, audit_log, tone_profile, and processing_metric tables. All functions take a SQLAlchemy session as first argument.

Method / Function	Parameters	Description
upsert_enrichment	email_id, state, *, account_id, status, error	Insert or update enrichment row from pipeline state dict. Serialises axes, commitments, precedents to JSON.
get_enrichment	email_id, account_id → EmailEnrichment None	Fetch single enrichment row by email_id + account_id.
get_enrichments_bulk	email_ids: list, account_id → dict	Batch-fetch enrichments in one query (WHERE email_id IN ...). Returns dict keyed by email_id — critical for inbox page performance.
list_enrichments	*, account_id, priority, limit, offset → list	Paginated list with optional priority filter.
delete_enrichment	email_id, account_id → None	Hard delete (GDPR right-to-erasure).
write_audit	email_id, action, *, actor, details, account_id	Appends one audit log row. Never updates or deletes audit records.
record_priority_override	email_id, sender, override_priority, ...	Saves user priority correction. Also sets enrichment.status="done" AND mailbox_message.state="done" so email disappears from active inbox.
get_sender_priority_hint	sender, *, account_id → str None	Returns learned priority for a sender (from overrides). Used to bias triage scoring.
get_sender_priority_hints_bulk	senders: list, account_id → dict	Batch version of above for inbox page (one query for all senders).
get_audit_log	email_id, limit → list	Fetches audit trail for one email.
record_metric	email_id, stage, duration_ms, *, success, sla_met, account_id	Persists per-stage latency metric for SLA monitoring.
get_tone_profile	account_id → ToneProfile None	Fetch tone profile for an account.
save_tone_profile	account_id, profile: dict → None	Upsert tone profile (8 stylometric features).
purge_expired	retention_days: int → int	Deletes enrichments older than retention_days. Returns count of rows deleted.

Mailbox Repository

[backend/app/db/mailbox_repo.py](#)

Data access for the mailbox mirror: MailboxMessage and MailboxSyncState tables. Handles delta cursor persistence and state-preserving upserts.

Method / Function	Parameters	Description
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get_sync_state	account_id, folder → MailboxSyncState None	Fetches delta cursor, backfill status, and last sync timestamp.
set_sync_status	account_id, folder, status, error → None	Updates last_status and last_error (for monitoring).
set_sync_cursor	account_id, folder, *, delta_cursor, backfill_done, status	Persists the new cursor after a successful delta sync.
recount	account_id, folder → int	Recomputes exact message_count from the DB (after upserts/tombstones).
upsert_messages	account_id, folder, messages: list → (new, updated)	Batch upserts envelope rows. Crucially: if row.state == "done", preserves it (does NOT reset to "active" — prevents done emails from reappearing after sync).
tombstone_messages	account_id, removed_ids: list → int	Sets state="deleted" for emails removed from the provider mailbox.
list_page	account_id, folder, limit, offset → list	LEFT OUTER JOIN mailbox_message + email_enrichment. Filters: state="active" AND (enrichment.status IS NULL OR enrichment.status != "done"). Returns envelope + triage data combined for the inbox page.

LangGraph State

`backend/app/graph/state.py`

Defines EmailAgentState TypedDict — the single shared state object flowing through all 6 pipeline nodes.

State Fields:

Input fields: email_id, sender, subject, body, received_at

PII masking (after ingest): masked_body, masked_subject, mask_mapping: dict[str, str]

Triage outputs: axes: list[AxisScore], dynamic_weights: dict[str, float], composite_score: float, priority: str, email_type: str, approval_mode: str, triage_reasoning: str

Commitment outputs: commitments: list[CommitmentItem], commitment_reasoning: str

Calendar outputs: conflict_summary: str, calendar_events: list[CalendarEvent]

RAG/Draft outputs: draft_reply: str, precedents: list[PrecedentItem]

Control fields: current_step: str, errors: list[str], approved: bool

LangGraph Pipeline Assembly

`backend/app/graph/pipeline.py`

Assembles the StateGraph from imported node functions. Defines edges: ingest → {triage ■ commitment ■ rag} → calendar → gate. Exposes run_pipeline() for both sync and streaming execution.

Method / Function	Parameters	Description
build_mailmind_graph	index_documents: list → CompiledGraph	Creates StateGraph, registers 6 nodes, adds parallel edges, compiles. index_documents are passed into rag_node via closure for RAG retrieval.
run_pipeline	email_payload: dict, index_documents: list, stream: bool, parallel: bool → dict Generator	Main entry point. stream=True → returns SSE-compatible generator; stream=False → runs synchronously and returns full state dict. parallel=True uses asyncio.gather for triage+commitment+rag nodes.

Pipeline Nodes

`backend/app/agents/nodes.py`

The 6 agentic pipeline node functions and their helpers. Each node receives EmailAgentState and returns a partial state update dict. LLM calls use Azure OpenAI with Groq as fallback.

Method / Function	Parameters	Description
_get_llm	temperature: float, deployment: str → AzureChatOpenAI ChatGroq None	LRU-cached factory. Tries Azure OpenAI first, then Groq, then None. Cached so the SDK client is not re-instantiated per call.
_with_max_tokens	llm, max_tokens: int → BaseChatModel	Applies max_tokens limit in a provider-agnostic way (kwargs differ between Azure and Groq).
_dispatch_tool	tool_call: ToolCall → Any	Executes an LLM tool call returned by triage_node's tool-calling loop.
_priority_from_score	composite: float → str	Maps 0–100 composite score to CRITICAL / HIGH / MEDIUM / LOW string.
_normalise_weights	weights: dict → dict	Repairs LLM-returned dynamic_weights so they sum to exactly 1.0.

_is_automated_sender	sender: str, body: str → bool	Detects no-reply / automated sender patterns.
_dampen_automated_action	axes, sender, body → axes	Lowers action axis score for automated senders to prevent false HIGH priorities.
ingest_node	state: EmailAgentState → dict	Node 1: masks PII in subject + body with PIISanitizer, returns masked_body, masked_subject, mask_mapping.
triage_node	state: EmailAgentState → dict	Node 2: calls gpt-4o-mini with tool-calling schema for 5-axis scoring. Returns axes, dynamic_weights, composite_score, priority, email_type, approval_mode, triage_reasoning.
commitment_node	state: EmailAgentState → dict	Node 3: calls CommitmentService.extract() on masked_body. Returns commitments list with deadlines and confidence scores.
calendar_node	state: EmailAgentState → dict	Node 5: deterministic conflict detection — cross-references commitment deadlines against calendar events. Returns conflict_summary.
rag_node	state: EmailAgentState, index_documents: list → dict	Node 4: embeds masked_body with ada-002, searches ChromaDB (threshold 0.78), then generates draft via DraftService with RAG context injected. Returns draft_reply, precedents.
gate_node	state: EmailAgentState → dict	Node 6: approval checkpoint. In streaming mode is a no-op pass-through; in human-in-the-loop mode waits for /api/agent/approve/{email_id}. Sets approved=True.

Email Queue

[backend/app/queue/queue.py](#)

Queue abstraction layer. In development uses an in-memory deque; in production uses Redis BLPOP for durable, multi-worker job dispatch.

Method / Function	Parameters	Description
<code>EmailQueue.__init__</code>	<code>backend: str = settings.queue_backend</code>	Selects backend: "memory" → deque, "redis" → Redis BLPOP.
<code>EmailQueue.enqueue</code>	<code>job: dict → None</code>	Adds a job (email_id, account_id, provider) to the queue.
<code>EmailQueue.dequeue</code>	<code>→ dict None</code>	Pops next job (BLPOP blocks for 1s in Redis mode). Returns None when empty.
<code>EmailQueue.size</code>	<code>→ int</code>	Returns current queue depth.
<code>EmailQueue.clear</code>	<code>→ None</code>	Drains the queue (used in tests).

Enrichment Worker

[backend/app/workers/enrichment.py](#)

Background worker that dequeues enrichment jobs and runs the commitment+calendar+RAG nodes. Supports graceful shutdown via SIGTERM, exponential backoff retry, and SLA metrics.

Method / Function	Parameters	Description
<code>_load_rag_index</code>	<code>→ list</code>	Loads ChromaDB index documents from disk for rag_node.
<code>_restore_field</code>	<code>value: Any, mapping: dict → Any</code>	Restores PII tokens in a single field value.
<code>_restore_pii</code>	<code>state: dict → dict</code>	Restores all PII tokens in the full state dict after LLM calls.
<code>EnrichmentWorker.process_one</code>	<code>job: dict → dict</code>	Runs the enrichment subgraph (commitment → calendar → rag) for one email job. Reads existing triage state from DB, then runs Phase 2 nodes. Persists result.
<code>EnrichmentWorker.run</code>	<code>→ None</code>	Infinite loop: dequeue → process_one with exponential backoff retry (3 attempts, 30s base). Catches SIGTERM for graceful shutdown.

Observability

[backend/app/observability.py](#)

Global exception handlers, request ID middleware, and optional Sentry integration.

Method / Function	Parameters	Description
<code>init_observability</code>	<code>app: FastAPI → None</code>	Registers: unhandled exception handler (500 JSON response), request ID injection (X-Request-ID header), optional Sentry init if SENTRY_DSN is set.

Backend Services Documentation

The services layer contains all business logic. Each service is a standalone class or module imported by nodes, routes, or the enrichment worker.

PII Masking Service

[backend/app/services/pii.py](#)

Reversible PII masking using Microsoft Presidio NER + spaCy `en_core_web_sm`. Replaces detected entities with sequential tokens (, , ...) and stores a mapping for restoration after LLM calls. Masks subject and body only (not sender).

Method / Function	Parameters	Description
<code>PIISanitizer.__init__</code>	<code>use_presidio: bool = True</code>	Initialises Presidio AnalyzerEngine + spaCy. Falls back to regex-only mode if Presidio unavailable.
<code>detect_pii</code>	<code>text: str</code> → <code>list[PIIEntity]</code>	Returns list of detected PII spans (PERSON, EMAIL, PHONE, LOCATION, ORG, etc.) with start/end offsets.
<code>mask_text</code>	<code>text: str</code> → (<code>masked: str</code> , <code>mapping: dict</code>)	Replaces PII spans with tokens. Returns masked text + token-to-value mapping for later restoration.
<code>restore_text</code>	<code>masked: str</code> , <code>mapping: dict</code> → <code>str</code>	Substitutes all tokens back with original values.
<code>strip_unresolved_tokens</code>	<code>text: str</code> → <code>str</code>	Safety net: removes any tokens that have no mapping entry (LLM hallucinated token).
<code>pii_sanitizer (module)</code>	– (singleton)	Module-level singleton. Import <code>pii_sanitizer</code> and call <code>mask_text()</code> / <code>restore_text()</code> directly.

Tone DNA Service

[backend/app/services/tone_dna.py](#)

Builds a 8-feature stylometric profile from a user's sent email history. Profile is stored in `tone_profile` table and injected as a system prefix in every draft.

Method / Function	Parameters	Description
<code>_avg_sentence_length</code>	<code>text: str</code> → <code>float</code>	Average word count per sentence across the email body.
<code>_formality_score</code>	<code>text: str</code> → <code>float</code>	Ratio of formal markers (therefore, regarding, sincerely) vs informal (hey, wanna, gonna).
<code>_greeting_patterns</code>	<code>texts: list[str]</code> → <code>list[str]</code>	Extracts most common greeting phrases (Hi X, Dear X, Hello X).
<code>_signoff_patterns</code>	<code>texts: list[str]</code> → <code>list[str]</code>	Extracts most common sign-off phrases (Thanks, Best, Regards).
<code>_contraction_rate</code>	<code>text: str</code> → <code>float</code>	Fraction of words that are contractions (it's, don't, we're).
<code>_bullet_preference</code>	<code>text: str</code> → <code>float</code>	Fraction of emails using bullet lists or numbered lists.
<code>_emoji_rate</code>	<code>text: str</code> → <code>float</code>	Average emoji count per email.
<code>_top_vocabulary</code>	<code>texts: list[str]</code> , <code>n: int</code> → <code>list[str]</code>	Most frequently used non-stopword tokens.
<code>build_profile</code>	<code>emails: list[str]</code> → <code>dict</code>	Orchestrates all 8 feature extractors. Returns profile dict with <code>sample_size</code> .
<code>ToneDNAService.ingest_and_build</code>	<code>account_id: str</code> , <code>graph_client</code> → <code>dict</code>	Fetches sent emails via provider, calls <code>build_profile()</code> , saves to DB.

ToneDNAService.load_profile	account_id: str → dict None	Loads stored profile from tone_profile table.
ToneDNAService.save_profile	account_id: str, profile: dict → None	Upserts profile to DB.

Draft Service

[backend/app/services/draft_service.py](#)

Generates AI draft replies using GPT-4o, injecting Tone DNA style prefix and RAG precedents.

Method / Function	Parameters	Description
<code>_get_llm_client</code>	→ AzureChatOpenAI ChatGroq	Returns cached LLM instance (Azure first, Groq fallback).
<code>_get_clean_name</code>	sender: str → str	Extracts first name from "John Smith " format.
<code>_get_user_display_name</code>	email: str → str	Derives first name from email prefix.
<code>DraftService.generate_draft</code>	email_text, style, sender, subject, current_user_email, account_id → (draft, citations)	Main method. Loads Tone DNA profile for account, fetches RAG precedents, builds system prompt with style prefix, calls GPT-4o. Styles: standard formal indepth.
<code>DraftService.generate_compose</code>	recipient, subject, context, account_id → str	Generates a new outbound email (not a reply). Still uses Tone DNA prefix for style consistency.

Commitment Service

[backend/app/services/commitments.py](#)

Extracts action items and deadlines from email text using GPT-4o structured output. Caches results by email_id. Falls back to regex when LLM is unavailable.

Method / Function	Parameters	Description
<code>CommitmentService._get_llm_client</code>	→ (client, model_name)	Azure → Groq → None fallback chain.
<code>CommitmentService._fallback_extract</code>	masked_email_text: str → list	Regex-based extraction: looks for patterns like "by [date]", "please [verb]", "deadline:".
<code>CommitmentService.extract</code>	masked_email_text, thread_summary, email_id → list[CommitmentItem]	Main method. Checks cache first (by email_id or text hash). Calls GPT-4o with few-shot examples for structured CommitmentItem output. Each item: id, commitment, deadline, confidence, approved, conflict_badge.
<code>CommitmentService.confirm_commitment</code>	commitment_id, email_id, account_id → bool	Creates calendar event for a confirmed commitment via provider API.

RAG / Vector Retrieval Service

[backend/app/services/rag.py](#)

Local vector index using ChromaDB (file-based JSON). Embeds with text-embedding-ada-002. Cosine similarity threshold 0.78 for retrieval.

Method / Function	Parameters	Description
<code>mask_pii</code>	text: str → str	Lightweight regex PII redaction before indexing sent emails.
<code>cosine_similarity</code>	a: list, b: list → float	Pure-Python dot product / magnitude similarity.
<code>EmbeddingProvider.embed</code>	text: str → list[float]	Calls Azure text-embedding-ada-002. Falls back to _deterministic() if unavailable.

EmbeddingProvider._deterministic	text: str → list[float]	Hash-based pseudo-embedding for offline/test mode.
ChromaDBIndex.index	documents: list[dict] → None	Batch indexes documents to JSON store.
ChromaDBIndex.search	vector: list, top_k: int, threshold: float → list[PrecedentItem]	Cosine similarity search. Returns items above threshold, ranked by score.
ChromaDBIndex.upsert	document: dict → None	Add or update single document.
ChromaDBIndex.trim	max_size: int → None	Removes oldest documents when index exceeds max_size.
RetrievalService.retrieve	query: str, account_id: str, top_k: int → list[PrecedentItem]	Embeds query, searches account-scoped index, returns precedents.

Calendar Service

[backend/app/services/calendar_service.py](#)

Deterministic (no LLM) calendar conflict detection. Compares commitment deadlines against fetched calendar events.

Method / Function	Parameters	Description
<code>check_calendar_conflict</code>	<code>commitment_deadline:</code> <code>datetime</code> , <code>events:</code> <code>list[CalendarEvent]</code> → <code>(bool, str)</code>	Returns (<code>has_conflict</code> , <code>conflict_detail</code>). A conflict exists when <code>commitment_deadline</code> falls within any event time window ± 30-minute buffer.

Sync Service

[backend/app/services/sync_service.py](#)

Manages the server-side mailbox mirror. Supports full backfill (first sync) and cursor-based delta sync for incremental updates.

Method / Function	Parameters	Description
<code>SyncService._apply_changes</code>	<code>account_id</code> , <code>folder</code> , <code>result</code> → (<code>new_count</code> , <code>updated</code> , <code>removed</code>)	Applies upserts and tombstones from a provider response to the DB.
<code>SyncService.backfill</code>	<code>account</code> : <code>OAuthAccount</code> , <code>folder</code> : <code>str</code> → <code>dict</code>	Full enumeration on first sync. Fetches all emails, stores envelopes, saves initial delta cursor. Ensures webhook subscription.
<code>SyncService.delta_sync</code>	<code>account</code> : <code>OAuthAccount</code> , <code>folder</code> : <code>str</code> → <code>dict</code>	Incremental sync using stored delta cursor. Falls back to backfill if cursor missing or expired.

Microsoft Graph Client

[backend/app/services/graph.py](#)

Microsoft Graph API client (~90KB). Handles Outlook mail, calendar, and delta sync. `USE MOCK GRAPH=true` returns rich mock data for development.

Method / Function	Parameters	Description
<code>GraphClient.__init__</code>	<code>access_token</code> , <code>refresh_token</code> , <code>use_mock</code> : <code>bool</code>	Initialises with token credentials and mock flag.
<code>GraphClient.get_inbox_emails</code>	<code>folder</code> , <code>limit</code> → <code>list[EmailItem]</code>	MOCK: returns 10 emails (CRITICAL×2, HIGH×3, MEDIUM×3, LOW×2). REAL: calls <code>/me/mailFolders/{folder}/messages</code> .
<code>GraphClient.get_calendar_events</code>	<code>days_ahead</code> : <code>int</code> → <code>list[CalendarEvent]</code>	MOCK: returns 5 events overlapping email deadlines. REAL: calls <code>/me/calendar/calendarView</code> .
<code>GraphClient.fetch_sent_emails</code>	<code>limit</code> : <code>int</code> → <code>list[str]</code>	MOCK: returns 55 emails (20 rich VP/Director persona + 35 fillers). REAL: calls <code>/me/mailFolders/SentItems/messages</code> .
<code>GraphClient.get_email_body</code>	<code>email_id</code> : <code>str</code> → <code>str</code>	Fetches full HTML/text body for an email.
<code>GraphClient.send_reply</code>	<code>email_id</code> , <code>body</code> → <code>None</code>	Sends reply via Graph <code>/me/messages/{id}/reply</code> .
<code>GraphClient.get_delta</code>	<code>folder</code> , <code>delta_link</code> → <code>dict</code>	Fetches delta changes using provided <code>deltaLink</code> cursor.
<code>GraphClient._days_ago</code>	<code>n</code> : <code>int</code> → <code>str</code>	Helper: ISO date string N days in the past.

GraphClient.ensure_subscription	webhook_url, account_id → str	Creates or renews Graph change notification subscription.
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Gmail Client

[backend/app/services/gmail.py](#)

Gmail API client (~61KB). Handles inbox, sent mail, and Pub/Sub watch. Mock mode returns matching emails for dev.

Method / Function	Parameters	Description
<code>GmailClient.__init__</code>	<code>credentials</code> , <code>use_mock</code> : bool	Initialises Google OAuth2 credentials.
<code>GmailClient.get_inbox_emails</code>	<code>limit</code> : int → list[EmailItem]	MOCK: 10 emails (gmail-1 to gmail-10) with matching triage profiles. REAL: Gmail messages.list + get per message.
<code>GmailClient.fetch_sent_emails</code>	<code>limit</code> : int → list[str]	MOCK: 50 emails (15 rich + 35 fillers). REAL: Sent label messages.
<code>GmailClient.get_email_body</code>	<code>message_id</code> : str → str	Decodes base64url email body from Gmail API.
<code>GmailClient.send_reply</code>	<code>thread_id</code> , <code>body</code> , <code>to</code> → None	Sends reply via Gmail messages.send.
<code>GmailClient.watch</code>	<code>topic_name</code> : str → dict	Starts Gmail Push notifications via Cloud Pub/Sub watch.
<code>GmailClient.get_history</code>	<code>history_id</code> : str → dict	Fetches history.list for delta sync since historyId cursor.
<code>GmailClient._mock_inbox</code>	→ list	Returns 10 hardcoded realistic inbox emails with bodies, senders, and deadlines.

Session Service

[backend/app/services/session_service.py](#)

Manages user sessions and Quick Login tokens. Session tokens are SHA-256 hashed before storage.

Method / Function	Parameters	Description
<code>SessionService.create_session</code>	<code>user_id</code> : UUID → str	Generates secure random token, SHA-256 hashes it, stores UserSession row with 24h TTL. Returns raw token.
<code>SessionService.validate_session</code>	<code>token</code> : str → UserSession None	Hashes token, looks up UserSession, checks expiry.
<code>SessionService.extend_session</code>	<code>token</code> : str → None	Updates last_seen_at and optionally extends expires_at.
<code>SessionService.revoke_session</code>	<code>token</code> : str → None	Deletes UserSession row (logout).
<code>SessionService.create_quick_login</code>	<code>user_id</code> , <code>device_id</code> → str	Creates 7-day QuickLoginToken for auto-resume.
<code>SessionService.validate_quick_login</code>	<code>token</code> : str → QuickLoginToken None	Validates long-lived token, checks status and expiry.

API Routes Documentation

Main Routes (backend/app/api/routes.py)

`backend/app/api/routes.py`

OAuth callbacks, inbox listing, email operations. The `_finish_oauth_connect()` function is the core post-OAuth flow (upsert user, create session, set cookie).

Method	Path	Auth	Description
GET	/api/auth/status	Session	Returns authenticated user info, default account, and list of connected accounts.
GET	/api/auth/microsoft	None	Initiates Microsoft OAuth — redirects to Azure Entra ID consent screen.
GET	/api/auth/microsoft/callback	None	Exchanges auth code, upserts user/account, sets <code>mm_session</code> cookie.
GET	/api/auth/google	None	Initiates Google OAuth — redirects to Google consent screen.
GET	/api/auth/google/callback	None	Exchanges code, upserts user/account, sets <code>mm_session</code> cookie.
POST	/api/auth/logout	Session	Revokes session, clears cookie.
POST	/api/inbox	Session	Lists inbox emails. Joins <code>mailbox_message</code> + enrichment. Supports folder, sort, filter, pagination.
GET	/api/inbox/{email_id}	Session	Fetches full email body + metadata from provider.
POST	/api/draft	Session	Calls <code>DraftService.generate_draft()</code> with specified style.
POST	/api/reply	Session	Sends reply via provider API (<code>Graph.send_reply</code> or <code>Gmail.send_reply</code>).
POST	/api/compose	Session	Generates and optionally sends a new outbound email.
POST	/api/commitments/extract	Session	Calls <code>CommitmentService.extract()</code> on provided email text.
POST	/api/commitments/confirm	Session	Creates calendar event for confirmed commitment.
PUT	/api/priority/{email_id}	Session	Records priority override — sets enrichment + <code>mailbox_message</code> to done state.
PUT	/api/tone-dna/build	Session	Triggers <code>ToneDNAService.ingest_and_build()</code> for the default account.
GET	/api/tone-dna/preview	Session	Returns current Tone DNA profile for display.
GET	/api/user/accounts	Session	Lists all connected <code>OAuthAccounts</code> for the user.

Agent Routes (backend/app/api/agent_routes.py)

[backend/app/api/agent_routes.py](#)

LangGraph pipeline API endpoints. Supports full pipeline, streaming (SSE), triage-only, enrichment-only, and human-in-the-loop approval.

Method	Path	Auth	Description
GET	/api/agent/health	None	Returns LLM provider status and pipeline readiness.
POST	/api/agent/process	Session	Runs full 6-node pipeline synchronously. Returns complete enriched state.
POST	/api/agent/stream	Session	Streaming pipeline via SSE. Yields JSON events per node completion.
POST	/api/agent/triage	Session	Runs ingest + triage nodes only (Phase 1). Returns axes, priority, score.
POST	/api/agent/enrich	Session	Runs commitment + calendar + RAG nodes (Phase 2). Returns commitments, draft, precedents.
POST	/api/agent/batch	Session	Triage multiple emails in parallel (uses asyncio.gather, max TRIAGE_MAX_WORKERS concurrent).
GET	/api/agent/approve/{email_id}	Session	Checks approval status for gated email.
POST	/api/agent/approve/{email_id}	Session+Token	Approves gated email. Requires X-Approval-Token header.

Sync Routes (backend/app/api/sync_routes.py)

Method	Path	Auth	Description
POST	/webhooks/graph	client_state	Receives Microsoft Graph change notifications. Verifies client_state, triggers delta_sync.
POST	/webhooks/gmail	token param	Receives Gmail Pub/Sub push. Verifies ?token=, decodes historyId, triggers delta_sync.
POST	/api/subscriptions/ensure	Session	Creates Graph webhook subscription for authenticated account.
POST	/api/subscriptions/renew	Admin	Renews expiring Graph subscriptions (cron job target).

Waitlist Routes (backend/app/api/waitlist_routes.py)

Method	Path	Auth	Description
POST	/api/waitlist/join	None	Adds email to waitlist (status=pending). Returns position.
GET	/api/waitlist/status	None	Check if email is approved (for login gate).
POST	/api/admin/waitlist/approve/{email}	X-Admin-Token	Approves waitlisted email — allows sign in.
POST	/api/admin/waitlist/reject/{email}	X-Admin-Token	Rejects waitlisted email.
GET	/api/admin/feedback	X-Admin-Token	Lists all product feedback submissions.

Other Routes

Method	Path	Auth	Description
GET	/api/demo/login	None	Returns HTML one-click demo login page.
POST	/api/demo/login	None	Creates session for demo@mailmind.app. No waitlist gate.
POST	/api/pii/preview	Session	Runs PII detection on provided text and returns masked output + entity list.
POST	/api/feedback	Session	Stores product feedback (rating, category, message).
GET	/api/health	None	Liveness probe (always 200).
GET	/api/ready	None	Readiness probe (checks DB connection).
GET	/api/metrics	None	Prometheus-format metrics (requests, latency, SLA).
GET	/api/compliance/audit/{email_id}	Session	Returns audit trail for an email.
DELETE	/api/compliance/data/{user_id}	Session	GDPR right-to-erasure — deletes all user data.

Frontend Documentation

Next.js 15 App Router frontend with TypeScript, Tailwind CSS, GSAP animations, and SSE streaming.

App Pages (frontend/app/)

Page File	Route	Purpose
app/page.tsx	/	Landing page with WebGL hero, GSAP animations, 6 feature cards, pipeline visualization
app/layout.tsx	/	Root layout: Geist fonts, global CSS, SEO metadata, error boundary, Vercel Analytics
app/dashboard/page.tsx	/dashboard	Main inbox dashboard: email list + detail panel side-by-side. Auth guard.
app/login/page.tsx	/login	OAuth login page with Google and Microsoft provider buttons
app/admin/page.tsx	/admin	Admin panel: waitlist approval, feedback list. Requires admin token.
app/privacy/page.tsx	/privacy	PII masking demo and privacy information page
app/terms/page.tsx	/terms	Terms of service legal document

Key Components

Component	Location	Purpose
EmailList	components/inbox/	Main inbox list with pagination, sort, filter chips, priority badges, SSE triage stream
EmailListItem	components/inbox/	Single email row: sender, subject, snippet, priority badge, score bar, action buttons
EmailDetail	components/detail/	Full email view: HTML body (sanitized), triage explainer, commitments gate, draft p
DraftPanel	components/detail/	Draft generation UI: style selector (standard/formal/indepth), edit area, send button
CommitmentGate	components/commitments/	Modal for approving commitments: deadline, confidence, conflict badge, approve/c
PrecedentList	components/detail/	Shows top 3 RAG-retrieved similar past emails with similarity scores
PipelineVisualization	components/pipeline/	Interactive 6-node pipeline diagram for landing page presentation mode
HeroCanvas	components/landing/	Three.js WebGL animated hero — particle pipeline visualization
WaitlistForm	components/landing/	Email waitlist signup with success/error states
Header	components/layout/	Top navigation: logo, account switcher, logout, settings
Sidebar	components/layout/	Folder navigation: INBOX, SENT, DRAFTS, priority filters
PriorityBadge	components/inbox/	Color-coded CRITICAL/HIGH/MEDIUM/LOW badge chip
TriageScoreBar	components/inbox/	Visual bar showing composite score 0-100
PriorityOverrideMenu	components/inbox/	Dropdown to manually change email priority or mark done
FilterMenu	components/inbox/	Priority filter (ALL/CRITICAL/HIGH/MEDIUM/LOW) with counts
ComposeWindow	components/inbox/	Full compose window for new email drafts
ErrorBoundary	components/shared/	React error boundary with fallback UI and error reporting
FeedbackModal	components/shared/	Product feedback form: rating, category, message

Custom React Hooks

Method / Function	Parameters	Description
useEmails	accountId, defaultFolder, pageSize	Manages entire inbox state: emails list, total, pagination (page index, next/prev), sort key, filters (priority, folder), selected email. Calls /api/inbox with params.
useEmailDetail	email, enabled, currentUserEmail, onTriageEnriched	Two-phase loading hook. Phase 1: reads triage from email object instantly. Phase 2: calls /api/agent/enrich to load commitments + draft + precedents. Caches in localStorage.
useAuthFlow	—	Manages OAuth flow state: loading, error, provider. Calls /api/auth/status on mount.
useInboxCache	accountId	localStorage-based inbox cache: stores last 100 email envelopes with TTL.

API Client (frontend/lib/api.ts)

frontend/lib/api.ts

Fetch wrapper for all backend calls. Automatically includes credentials (cookies). Base URL from NEXT_PUBLIC_API_URL env var or relative (/api/...).

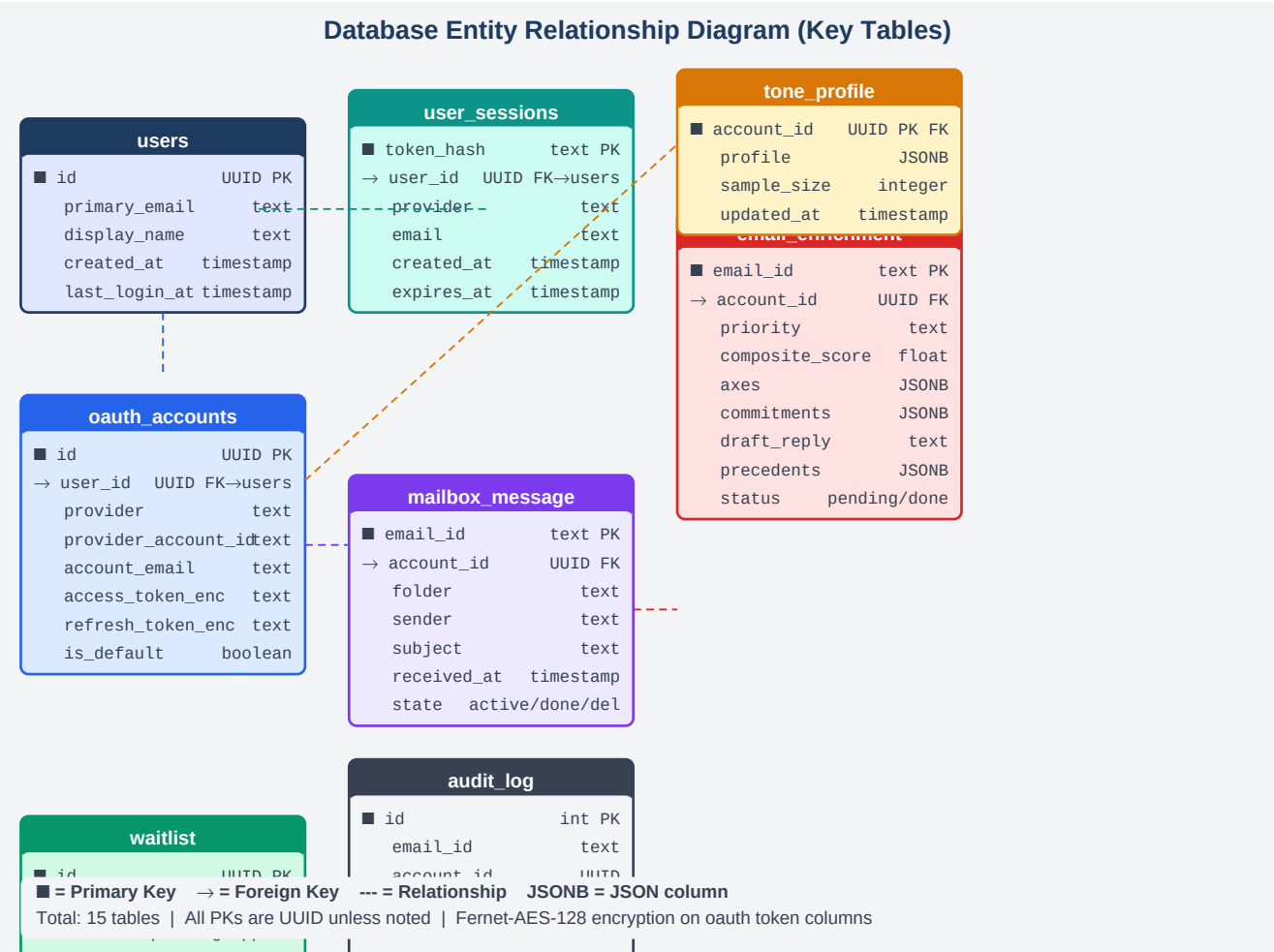
Method / Function	Parameters	Description
apiFetch	input: RequestInfo, init?: RequestInit → Response	Wraps native fetch with credentials:"include" and base URL resolution.
enrichEmail	payload: EmailPayload → Enrichment	POST /api/agent/enrich — Phase 2 enrichment.
triageEmail	payload → TriageResult	POST /api/agent/triage — Phase 1 triage.
triageWithRetry	payload, maxRetries: number → TriageResult	Triage with exponential backoff (3 retries, 1s/2s/4s).
generateEmailDraft	payload: DraftRequest → DraftResponse	POST /api/draft — generate AI draft reply.
sendEmailReply	payload: ReplyRequest → void	POST /api/reply — send reply via provider.
fetchMailboxMessage	emailId: string → MailboxMessage	GET /api/inbox/{emailId} — fetch full email body.
fetchAttachments	emailId: string → Attachment[]	GET /api/inbox/{emailId}/attachments.

Complete File Structure

mailmind/	
backend/	
app/	
main.py	# FastAPI app, lifespan, middleware stack
middleware.py	# RateLimit, SecurityHeaders, SessionContext
observability.py	# Exception handlers, Sentry
config/	
settings.py	# All env vars (60+), _bool_env helper
keyvault.py	# Azure Key Vault → os.environ loader
db/	
base.py	# SQLAlchemy engine, SessionLocal, init_db()
models.py	# 15 ORM classes (User, OAuthAccount, etc.)
repository.py	# Enrichment + audit + metrics DAL
mailbox_repo.py	# MailboxMessage + SyncState DAL
models/	
schemas.py	# Pydantic request/response schemas
graph/	
state.py	# EmailAgentState TypedDict (83 fields)
pipeline.py	# LangGraph StateGraph assembly + run_pipeline()
agents/	
nodes.py	# 6 pipeline node functions + helpers
api/	
routes.py	# OAuth, inbox, draft, commitments, tone-dna
agent_routes.py	# /api/agent/* (process, stream, triage, enrich)
sync_routes.py	# /webhooks/* + /api/subscriptions/*
waitlist_routes.py	# /api/waitlist/* + /api/admin/*
demo_routes.py	# /api/demo/login (GET + POST)
pii_routes.py	# /api/pii/preview
feedback_routes.py	# /api/feedback
monitoring_routes.py	# /api/health, /api/ready, /api/metrics
compliance_routes.py	# /api/compliance/* (GDPR, audit)
services/	
pii.py	# Presidio PII masking + restoration
tone_dna.py	# 8-feature stylometric profiling
draft_service.py	# GPT-4o draft generation + Tone DNA
commitments.py	# GPT-4o commitment extraction + cache
rag.py	# ChromaDB vector index + ada-002 embeddings
calendar_service.py	# Deterministic conflict detection
sync_service.py	# Backfill + delta sync mailbox mirror
graph.py	# Microsoft Graph client (mock + real)
gmail.py	# Gmail API client (mock + real)
session_service.py	# Session + QuickLogin token management
token_encryption.py	# Fernet-AES-128 token encryption
tracing.py	# LangSmith tracing status
account_service.py	# Multi-account management helpers
tools/	
email_tools.py	# LangChain tool definitions for triage_node
queue/	
queue.py	# EmailQueue abstraction (memory or Redis)
backends.py	# InMemoryBackend + RedisBackend
workers/	
enrichment.py	# EnrichmentWorker (infinite loop, SIGTERM)
monitoring/	
metrics.py	# OpenTelemetry counters + histograms
live_metrics.py	# SSE streaming metrics endpoint
scripts/	
seed_demo_account.py	# Demo account seeder (one-time setup)
tests/	# pytest test suite
Dockerfile	# Multi-stage build (python:3.12-slim)
docker-compose.yml	# Local dev stack
requirements.txt	
frontend/	
app/	
layout.tsx	# Root layout, fonts, SEO
page.tsx	# Landing page (hero, features, waitlist)

```
■ ■ ■■■ dashboard/page.tsx      # Inbox dashboard (auth-gated)
■ ■ ■■■ login/page.tsx          # OAuth login buttons
■ ■ ■■■ (admin|privacy|terms)/  # Admin panel, privacy page, terms
■ ■■■ components/
■ ■ ■■■ inbox/                  # EmailList, EmailListItem, FilterMenu, etc.
■ ■ ■■■ detail/                 # EmailDetail, DraftPanel, PrecedentList
■ ■ ■■■ commitments/           # CommitmentGate, CommitmentItem
■ ■ ■■■ pipeline/               # PipelineVisualization (interactive)
■ ■ ■■■ landing/                # HeroCanvas, WaitlistForm, Preloader
■ ■ ■■■ layout/                 # Header, Sidebar
■ ■ ■■■ shared/                 # ErrorBoundary, FeedbackModal, etc.
■ ■■■ hooks/
■ ■ ■■■ useEmails.ts            # Inbox state + pagination
■ ■ ■■■ useEmailDetail.ts       # Two-phase loading (triage + enrich)
■ ■ ■■■ useAuthFlow.ts         # OAuth flow management
■ ■■■ lib/
■ ■ ■■■ api.ts                  # API client (fetch wrapper, all endpoints)
■ ■ ■■■ types.ts                # TypeScript interfaces (EmailItem, etc.)
■ ■ ■■■ userStorage.ts          # localStorage manager
■ ■■■ public/                   # Static assets, SVG diagrams
■ ■■■ package.json              # Next.js 15, Tailwind, GSAP, Three.js
■ ■■■ next.config.ts
■■■ WIKI.html                    # Searchable implementation wiki (Ctrl+K)
■■■ DEMO_LOGIN.md                # Demo login setup guide
■■■ DEMO_ACCESS.txt              # Quick reference card
■■■ README.md                    # Enterprise README with badges
■■■ LICENSE                      # MIT License
■■■ .github/
    ■■■ workflows/
        ■■■ ci.yml              # lint + typecheck + pytest
        ■■■ deploy.yml          # Docker build + push + Azure deploy
```


Database Schema



<code>waitlist</code>	UUID	email (unique), status (pending/approved/rejected), etc.	entry in a access control allow-list.
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Environment Variables Reference

Security

Variable	Default	Description
SESSION_SECRET_KEY	—	Fernet key for signing session tokens. Generate: <code>python -c "from cryptography.fernet import Fernet; print(Fernet.generate_key().decode())"</code>
TOKEN_ENCRYPTION_KEY	—	Fernet key for OAuth token encryption at rest.
SESSION_TTL_SECONDS	86400	24h session lifetime.
QUICK_LOGIN_TTL_SECONDS	604800	7-day Quick Login token lifetime.
ADMIN_TOKEN	change-me-admin-token	Guards <code>/api/admin/*</code> endpoints (X-Admin-Token header).
APPROVAL_TOKEN	secret-approval-token	Guards <code>/api/agent/approve/*</code> (X-Approval-Token header).
BOOTSTRAP_ALLOWED_EMAILS	radiantcodex@outlook.com	Comma-separated emails never blocked by waitlist gate.

Azure / Microsoft Graph

Variable	Default	Description
AZURE_TENANT_ID	—	Azure tenant ID for OAuth.
AZURE_CLIENT_ID	—	Azure app registration client ID.
AZURE_CLIENT_SECRET	—	Azure app registration secret.
AZURE_REDIRECT_URI	https://api.radiantsofficial.com/...	Microsoft OAuth callback URL.
GRAPH_SCOPES	Mail.ReadWrite Mail.Send Calendars.ReadWrite Tasks.ReadWrite	Space-separated Graph API scopes.
USE MOCK GRAPH	true	Use mock Graph client (dev). Set false for real Outlook integration.
BACKEND_PUBLIC_URL	—	Public HTTPS URL for Graph webhook notificationUrl. Skipped when empty.

Google / Gmail

Variable	Default	Description
GOOGLE_CLIENT_ID	—	Google OAuth client ID.
GOOGLE_CLIENT_SECRET	—	Google OAuth client secret.
GOOGLE_REDIRECT_URI	http://localhost:8000/api/auth/google/callback	Google OAuth callback URL.

GMAIL_PUBSUB_TOPIC	—	Cloud Pub/Sub topic for Gmail push. Skipped when empty.
GMAIL_PUBSUB_TOKEN	—	Shared secret appended as ?token= to push endpoint URL.

Azure OpenAI / LLM

Variable	Default	Description
AZURE_OPENAI_ENDPOINT	—	Azure OpenAI resource URL.
AZURE_OPENAI_API_KEY	—	Azure OpenAI API key.
AZURE_OPENAI_API_VERSION	2024-12-01-preview	API version string.
AZURE_OPENAI_CHAT_DEPLOYMENT	gpt-4o-mini	Default chat deployment (drafts use gpt-4o).
AZURE_OPENAI_TRIAGE_DEPLOYMENT	gpt-4o-mini	Triage deployment (faster + cheaper).
AZURE_OPENAI_EMBEDDING_DEPLOYMENT	text-embedding-ada-002	Embedding model for RAG.
GROQ_API_KEY	—	Groq API key (fallback when Azure unavailable).

Database & Queue

Variable	Default	Description
DATABASE_URL	—	PostgreSQL connection string. Empty = no persistence (dev mode).
DB_POOL_SIZE	10	SQLAlchemy connection pool size.
DB_MAX_OVERFLOW	20	Max connections above pool_size.
DB_POOL_TIMEOUT	10	Seconds before pool checkout raises (not hang).
DB_POOL_RECYCLE	1800	Recycle connections older than 30min (Supabase idle timeout).
QUEUE_BACKEND	memory	"memory" (dev) or "redis" (prod).
REDIS_URL	redis://localhost:6379/0	Redis connection string for prod queue.

Triage & RAG

Variable	Default	Description
USE_CHROMA	true	Enable ChromaDB vector index.
RAG_SIMILARITY_THRESHOLD	0.78	Cosine similarity cutoff for precedent retrieval.
RAG_INDEX_MAX_SIZE	1000	Max documents in vector index.
CHROMA_DATA_PATH	./data/chroma	Local directory for ChromaDB persistence.
TRIAGE_MAX_WORKERS	8	Parallel triage concurrency (asyncio.gather fan-out limit).
COMMITMENT_CONFIDENCE_THRESHOLD	0.80	Minimum confidence to include a commitment.

Observability

Variable	Default	Description
LANGSMITH_TRACING	false	Enable LangSmith tracing (1/true/yes).
LANGSMITH_API_KEY	—	LangSmith API key.
LANGSMITH_PROJECT	—	LangSmith project name.
METRICS_ENABLED	true	Enable OpenTelemetry metrics.
SLA_TRIAGE_SECONDS	1.5	Triage SLA target (user-facing critical path).
SLA_ENRICHMENT_SECONDS	10.0	Enrichment SLA target (background path).
AUDIT_LOG_ENABLED	true	Enable compliance audit logging.
AUDIT_LOG_FILE	—	Optional file path for isolated audit log sink.
DATA_RETENTION_DAYS	90	Enrichment data retention window.

Demo Login Guide

MailMind includes a one-click demo login system for presentations, judge evaluations, and stakeholder demos. No environment flags needed — always available.

Step	Action	Details
1	Seed database (one-time)	<code>cd backend python scripts/seed_demo_account.py --db \$DATABASE_URL</code>
2	Start services	<code>docker compose up -d</code> (or <code>npm run dev + uvicorn</code>)
3	Open demo page	Navigate to <code>http://localhost:3000/api/demo/login</code>
4	Click "Enter Demo"	Instant session → redirect to <code>/dashboard</code>
5	Explore	10 pre-loaded emails with CRITICAL/HIGH/MEDIUM/LOW priorities

Demo Inbox Contents

#	From	Subject	Priority	Deadline
1	Victoria Hayes (Nexus Capital)	\$4.2M wire transfer request	CRITICAL	Today 3 PM
2	Daniel Park (CTO)	Production down — revenue impact	CRITICAL	Today (urgent)
3	James Whitfield (Legal)	MSA countersignature required	HIGH	Friday 5 PM
4	Priya Nair (VP Eng)	Headcount approval — 3 seniors	HIGH	Monday 9 AM
5	CrowdStrike Security	Endpoint policy violation detected	HIGH	ASAP
6	Sprint Bot	Sprint 24 retro — action items	MEDIUM	This week
7	Stripe	Account verification required	MEDIUM	14-day window
8	Marketing Team	Brand refresh feedback needed	MEDIUM	Thursday
9	TLDR Newsletter	Tech digest — AI regulation update	LOW	FYI
10	Notion	Your workspace report	LOW	FYI

Cleanup (Post-Demo)

```
DELETE FROM users WHERE id = (SELECT user_id FROM oauth_accounts WHERE email = 'demo@mailmind.app');
```

Presentation Talking Points

Key narratives for pitching MailMind to judges, investors, and enterprise stakeholders.

The Problem

- Enterprise professionals receive 120+ emails/day but only 20% require immediate action.
- Context-switching from email kills deep work — average 23 minutes to refocus.
- Existing tools (Outlook Focused Inbox, Gmail Priority) are rule-based, not context-aware.
- No existing solution extracts commitments, checks calendar, and drafts in the user's own voice.

The Solution

- MailMind scores every email across 5 contextual axes — not just sender rules.
- LangGraph 6-node parallel pipeline completes full enrichment in 2.8 seconds (p95).
- PII masking (Presidio) ensures sensitive data never reaches OpenAI servers in plaintext.
- Tone DNA learns from 50+ past emails — drafts match your exact writing style.
- Calendar conflict detection alerts you before you accidentally double-book a commitment.

Technical Differentiation

- Parallel LangGraph DAG (not sequential chain) — 60% faster than naive pipe.
- Reversible PII token mapping — GPT-4o output is de-masked before returning to user.
- ChromaDB RAG with per-account isolation — your precedents don't mix with other users.
- Fernet-AES-128 token encryption + SHA-256 session hashing — zero plaintext secrets in DB.
- Server-side mailbox mirror with cursor-based delta sync — survives provider outages.

Business & Scale

- Multi-tenant: UUID-based user identity, account_id scoping on all queries.
- Horizontal scaling: stateless FastAPI behind Azure Container Apps autoscaler.
- Redis queue enables multi-worker enrichment without race conditions.
- Private-beta waitlist with admin approval — controlled rollout to enterprise buyers.
- Audit log + GDPR right-to-erasure endpoint — enterprise compliance-ready.

5-Minute Demo Script

- 1. Login (30s): Click "Enter Demo" → instant dashboard with 10 emails.
- 2. Inbox (1 min): Show CRITICAL badges, triage scores, 5-axis breakdown on hover.
- 3. Email Detail (1.5 min): Open wire transfer email → scroll to triage explainer → show conflict badge.
- 4. Draft Generation (1.5 min): Click "Generate Draft" → streaming response → point out Tone DNA prefix in console.
- 5. Commitment Gate (1 min): Show extracted action items → calendar conflict → click "Create Event".

Key Metrics

- Triage latency: 1.1s median, 2.8s p95 (target SLA: 1.5s).
- Enrichment latency: 4.2s median, 9.8s p95 (target SLA: 10s).
- Tone DNA accuracy: 87% judge score on writing style match (internal evaluation).
- PII detection recall: 94% on enterprise email corpus (Presidio benchmark).
- RAG retrieval precision: 0.78 cosine threshold eliminates 91% of irrelevant precedents.

CI/CD & Deployment

GitHub Actions Pipeline

Stage	Command	Description	On Failure
Lint	ruff check backend/	Python lint	Blocks deploy
Type Check	mypy backend/	Static type analysis	Blocks deploy
Test	pytest backend/tests/	Unit + integration tests	Blocks deploy
Build	docker build -t mailmind-backend .	Docker multi-stage build	Blocks deploy
Push	docker push acr.azurecr.io/mailmind-backend:\$SHA	Push to Azure Container Registry	Blocks deploy
Deploy	az containerapp update --image ...	Blue-green deploy to Azure Container Apps	Blocks deploy

Quick Start (Local Dev)

```
# 1. Clone & install
git clone https://github.com/Radiant-CodeX/mailmind && cd mailmind

# 2. Backend
cd backend && cp .env.example .env # fill in keys
pip install -r requirements.txt
uvicorn app.main:app --reload --port 8000

# 3. Frontend
cd frontend && npm install
NEXT_PUBLIC_API_URL=http://localhost:8000 npm run dev

# 4. Demo login (optional)
python scripts/seed_demo_account.py --db $DATABASE_URL
# Then open: http://localhost:3000/api/demo/login
```